S.T. Yau High School Science Award

Research Report

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Title of Research Report

The Research and Design of Bionic Beaver Amphibious Robot with Flippers and Tails

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Michael Junming Huang

Abstract

Nowadays, robots are widely used in underwater rescuing missions, yet few existing systems can operate effectively both on land and in water. Amphibious locomotion would enable robots not only to patrol on land but also to provide timely help in aquatic emergencies. Current design, track-fin driven bionic amphibious robot, suffer from structural complexity, as both tracks and fins require independent drive systems, increasing potential failure points[17]. In this research, beaver [8] becomes the inspiration of the bionic amphibious robot due to its highly motive terrestrial and aquatic locomotion. Unlike track-fin designs, our robot achieves locomotion in both water and on land solely through flippers and tail mechanisms, simplifying the actuation system while enhancing stability and adaptability. By analyzing the anatomical structure and the moving patterns of beaver [8]s' hind legs and flippers, a robotic system is constructed with waterproof joints, 3D-printed modular components, and intelligent sensor integrations. With a reinforcement learning [1][2][3][4][5] algorithm (PPO [1][3][4]) being applied, it helps to improve the stability and the efficiency of the robot, which makes working both in aquatic and terrestrial environments possible for the robot. With novel innovations on motion modeling, inverse kinematics, and physical testing, the prototype of the robot shows significant improvements in amphibious mobility in experiment, and breaks the convention of traditional robotic systems. Overall, by combining the bionic inspiration and the developed algorithms, the robots are designed with flippers, tails and certain crucial joints, which enhances its mobility and stability while moving on land and swimming in water, thus contributing to its amphibious locomotion.

Keywords: Amphibious Robotics | Bionic Design | Reinforcement Learning (PPO [1][3][4]) | Hydrodynamic Stability | Beaver Inspired Locomotion | Motion Control | Underwater Robotics

Acknowledgement

To begin with, I would like to express my most genuine gratitude to my mentors for their guidance about the fundamentals of robotics that have been used in this project, including mechanical design, actuation systems, control and motion planning, and simulation and testing. Inspired by the drowning mortality and current underwater robots, with the constructive suggestions from the tutors, the topic regarding the bionic beaver [8] amphibious robot. With tutors' assistance, I designed the mechanical structure and acquired the basic data of the robot. By learning kinematics and conducting force analysis on the robot, I had calculated and analysed the force on every joint of the robot in order to make the structural design more stable and logical. Leading by the teachers, I had applied the PPO [1][3][4] reinforcement learning [1][2][3][4][5] on the robot. With mentors' help on conducting the experiment regarding the mobility and the functionality of the bionic robot, we had made analysis and comparisons about the optimal morphology for the robot. Finally, my tutor guided me through the structure of this essay, and I was responsible for all the contents regarding this topic.

Through this project, one of the most challenging parts is the set up part for the PPO [1][3][4] reinforcement learning [1][2][3][4][5]. The installation of Ubuntu is struggling, as the ubuntu system is unable to detect the driver of my computer. With the guidance from the tutors, the problem is resolved by installing wls on windows instead of using dual-boot configuration. Beyond operating system issues, other challenges in setting up reinforcement learning [1][2][3][4][5] such as configuring the necessary Python dependencies and libraries, ensuring GPU compatibility for faster training, handling version mismatches between packages, and managing computational resource limitations, all of which are solved by conducting online research about the solutions regarding each of the challenges mentioned. Additionally, correctly integrating the environment with RL frameworks and debugging errors during initial runs can be time-consuming and require careful troubleshooting with great patience.

Therefore, overall, I felt really grateful about the knowledge my mentors have taught to me through this project, also the encouragement and invaluable guidance they have provided whenever encountering challenges.

Declaration of Academic Integrity

The participating team declares that the paper submitted is comprised of original research and results obtained under the guidance of the instructor. To the team's best knowledge, the paper does not contain research results, published or not, from a person who is not a team member, except for the content listed in the references and the acknowledgment. If there is any misinformation, we are willing to take all the related responsibilities.

Names of team members：Michael Huang

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Date 08/23/2025

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1. Introduction

The application of robots in aquatic rescues has gradually become a tendency. According to WHO(The World Health Organization [13]), there are around 300 000 annual drowning deaths worldwide. In order to alleviate the deaths caused by drowning, underwater robots are utilized. For instance, the FIFISH Underwater Rescue Robot [14] and EMILY [15]. For the former robot, it is powered by a compact lithium battery and is equipped with AI-based target locking, diver tracking, and real-time video feedback. It is widely used in underwater bomb disposal, underwater rescues and disaster rescues. For the latter one, it's a remote controlled surf-size robot and it is dropped into water manually by staff. Even though these robots contribute to all kinds of rescue missions, none of them are amphibious. Current design, track-fin driven bionic amphibious robot, suffer from structural complexity, as both tracks and fins require independent drive systems, increasing potential failure points[17]. Therefore, by designing the bionic beaver [8] amphibious robot, it can achieves locomotion in both water and on land solely through flippers and tail mechanisms, simplifying the actuation system while enhancing stability and adaptability. Most importantly, it can facilitate the development of robotics in underwater rescues and help to decrease the annual drowning mortality worldwide.

The bionic beaver [8] amphibious robot is an innovative robotic design that dives into the working mechanics of beaver [8]'s anatomical structure. Currently, there are a few studies regarding the swimming control of the amphibious bionic robot. As its motion control and hydrodynamic modeling is difficult to achieve stable motion control over the webbed underwater bionic robots. However, through this research, the stable control over the flippers of the bionic beaver [8] amphibious robot is going to be achieved by combining motion modeling and inverse kinematics with reinforcement learning [1][2][3][4][5] algorithms.

Therefore, in this essay, it's going to inquire the design of a bipedal swimming robot structure modeled after a beaver [8] and prototyping; the developing process of the RL-based motion control algorithm for the beaver [8]-inspired swimming robot, which covers action/observation spaces, reward function design, control programming, and prototype debugging; and the validation of adaptability in static and dynamic aquatic environments with experimental testing.

2. Methods

2.1. Beaver-Like Mechanical Design

In order to make a bionic beaver [8] robot, it's significant to examine the anatomical structure of the beaver [8]. To begin with, a beaver [8] has 2 forelimbs and 2 hind limbs. The forelimbs(Figure 1.) of beaver [8]s are short and muscular to help beaver [8]s to support itself while moving on land. Additionally, it doesn't have any webbed skin between its toes. Structurally, 4 joints, shoulder joint, elbow joint, wrist joint, and digit joints made up the forelimbs.

Figure 1. The Forelimb of the Beaver

Beaver's hind limb structure and its characteristic webbed hind feet are displayed in Figure 2. Its hind limb is mainly composed of three joints, hip, knee and ankle. The hip connects with the main body of the beaver [8], and the ankle connects with the webbed hind feet, which helps to control the movement of the beaver [8]. On its hind feet, there are 4 toes per each foot, and they are connected by the flexible webbed skin between them. With the physical adjustment of the joints on the hind feet, the beaver [8] is capable of controlling the contraction and extension of the webbed skin, which helps to increase or decrease the water resistance beaver [8] encounters while swimming. Therefore, it assists the beaver [8] to accelerate or decelerate in aquatic environments.

Figure 2. Hind Limbs and Hind Feet

2.1.1. Forelimbs Mechanical Structure Design

In Figure 3, it shows the design of the forelimbs. Even though the forelimbs are composed of 4 joints, the shoulder joints and digit joints are ignored in the design as the forward-backward motion of the robot is satisfied by only having the other two joints. Each joint on each forelimb is controlled by one of the two waterproofed servos to support its movement.

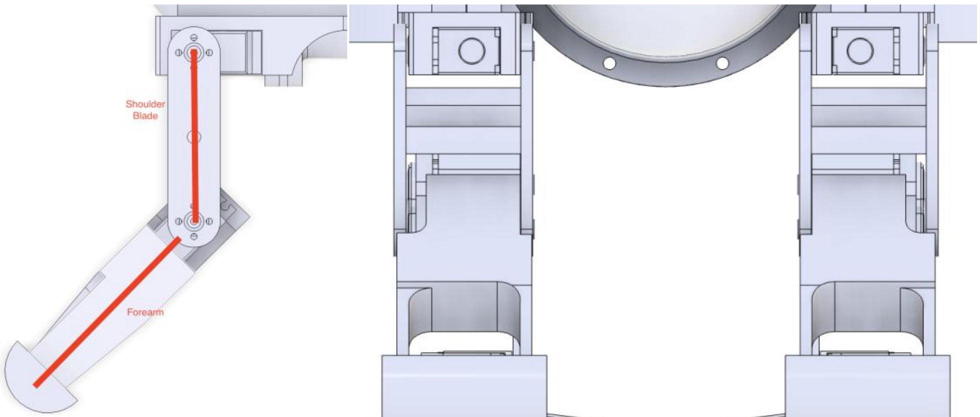

Figure 3. The Mechanical Design of the Forelimbs(left view & front view)

2.1.2. Hind limbs Mechanical Structure Design

For the hind limbs (Figure 4), there are three joints, and each of them are mimicked by using water proofed servos. The hind feet of the robot is designed in a sector shape with three toes in order to generalize the basic structure of beaver [8]'s hind feet. Additionally, the tail is designed to mimic the beaver [8]'s elongated oval shape and is driven by an independent servo motor to oscillate vertically, which helps to stabilize the robot's pitch during swimming and enhances its swimming speed.

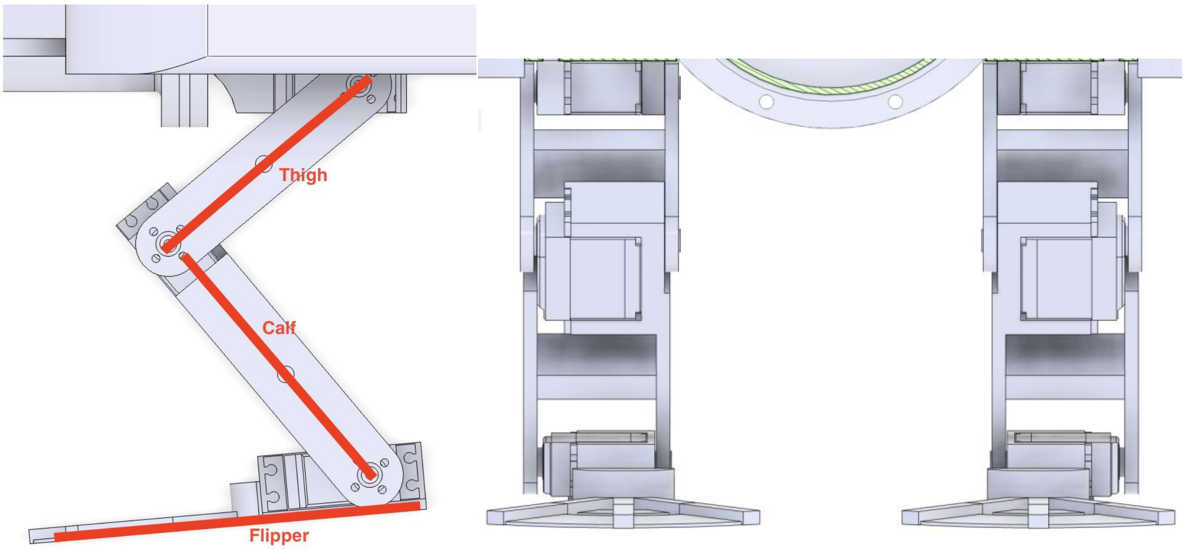

Figure 4. The Mechanical Design of The Hind Limbs(left view & front view)

2.1.3. Sealed Cabin Design

The cabin is the place where all the hardwares such as the main board, servo drivers, and high precision altitude sensor are settled. Therefore, within the cabin there are a lot of wires connecting to the external ultrasonic sensor and the servos that assist the locomotion of the robot. Figure 5 shows the internal distribution of different hardwares within the cabin. As all the main operational hardware is inside the cabin, the waterproof capability of the cabin needs to be guaranteed. In Figure 6, the sealed cabin is a cylindrical structure, which makes the aquatic locomotion of the robot to be less susceptible to the water resistance. The rubber mat and sealed joints are applied to make it more waterproof, therefore hindering the water from leaking into the cabin from the connections between the cabin and other parts of the robots. In addition, by using the sealed joint, it makes wires to connect to the external devices without water leaking into the cabin.

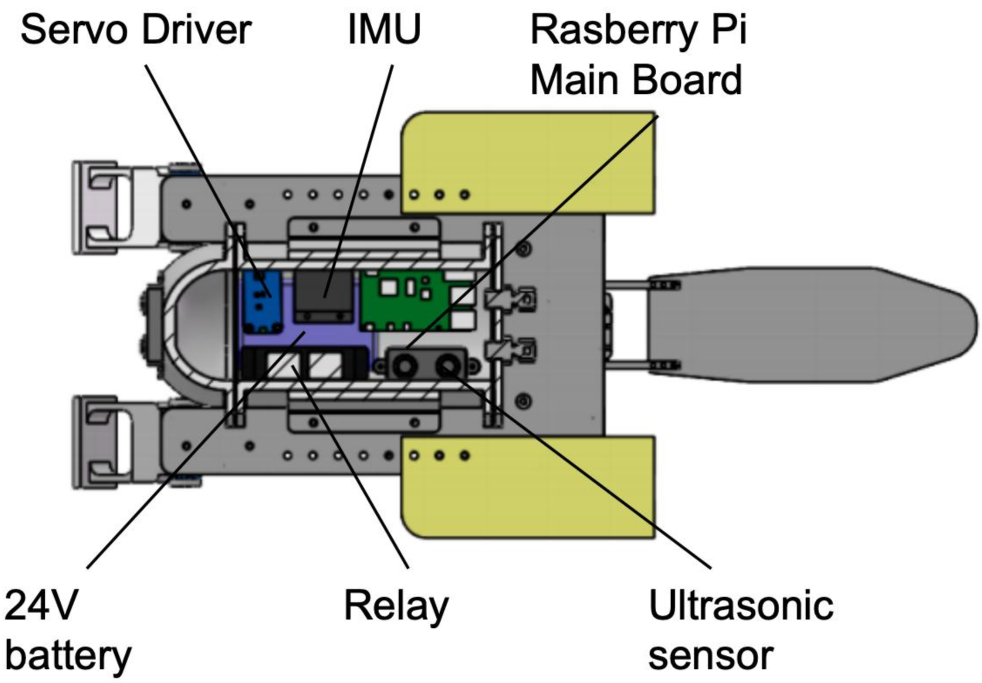

Figure 5. The Internal Distribution of The Waterproof Cabin

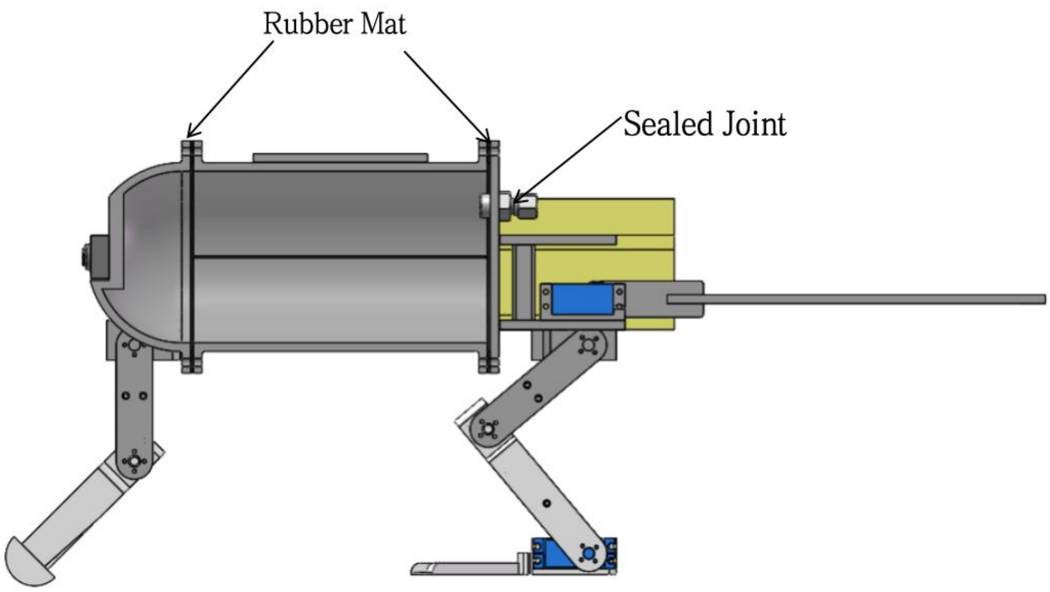

Figure 6. The Waterproof Measures on the Sealed Cabin

2.1.4. Tail Structure Design

The tail is an important part of this bionic beaver [8] robot, as it’s not only an unique part that the beaver [8] has but also acts as a stabilizer while the beaver [8] is swimming. Incorporating the tail improves the robot’s posture and stability, enhancing the locomotive efficiency. The tail is in a long oval shape that is similar to the one of the beaver [8]'s. It is connected to the main body by a waterproof servo which can control the tail to flip upside down while the robot is swimming in the water, as it can counteract with the water flow effectively. The tail is shown in Figure 7.

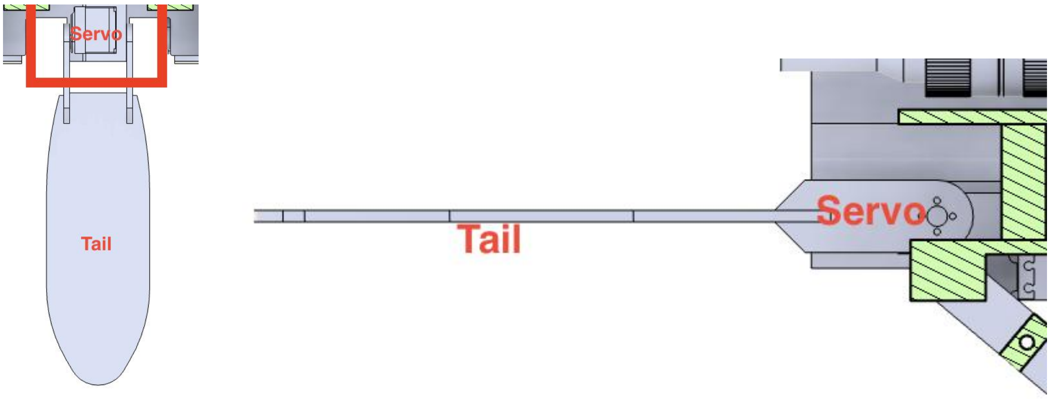

Figure 7. The Tail Structure Design of the Bionic Robot

2.1.5. The Overall Design of the Robot

As shown in Figure 8, the structure of the beaver [8]-inspired amphibious robot is mainly composed of left and right hind limbs, tail, flippers, body, sealed cabin, buoyancy block, etc. The left and right hind limbs are installed symmetrically to prevent rollover, and the hip joint motor, knee joint, and ankle joint are connected to the corresponding parts respectively. The sealed cabin is 3D printed, waterproofed with silicone pads and bolted. The upper and lower layers of the cabin bracket are respectively fixed with the main control chip, attitude sensor and motor control module, an external power supply, and a single motor drives the long tail.

In Figure 9, the form displays the general information of the robot, including the measurements regarding the mass, the length of the tail, the flippers, and the hind limbs of the robot.

Figure 8. 3D Model design

Parameters	Descriptions	Values
M	Mass	3.65kg
L_t	Length of the Tail	222mm
L_w	Length of the Flippers	75mm
L_h	Length of the Hind Limbs	170mm

Figure 9. The General Information of the Robots

2.2. The Hardware System Design

The hardware system design is displayed in Figure 10, the system mainly consists of a main control chip, external sensor modules, a motor driver module, and a power voltage divider module. The main control chip is responsible for algorithm training and outputting control signals, and it also supports remote control of the robot via SSH. The external sensor modules are used to monitor the surrounding environment and the robot’s internal status. The motor driver module receives control signals from the main chip via the I2C interface to control multiple motors. The power supply module provides multiple voltage outputs to ensure stable operation of all modules.

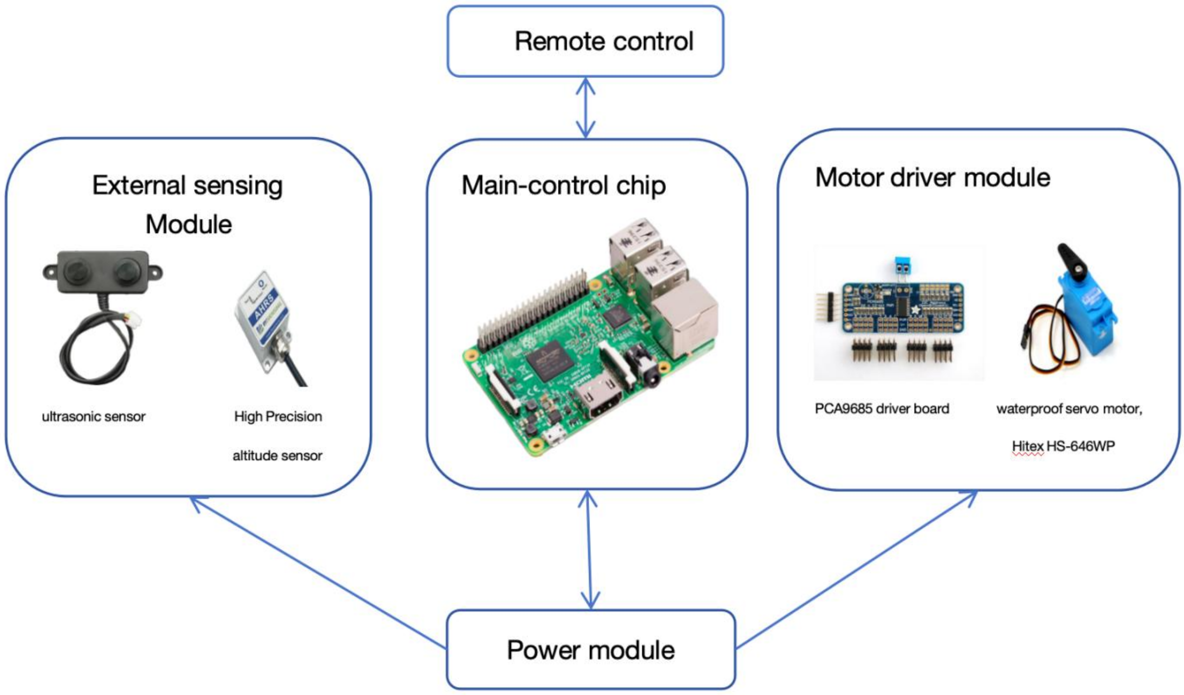

Figure 10. Hardware System Design

2.2.1. The Main Control Chip

The main control chip is mainly used to perform iterative computations and neural network inference to generate motor control signals, and to support remote access to the onboard system via SSH for controlling the robot's movement. Therefore, it has a high requirement for the main control chip, and the chip needs to have common functions of embedded chips include GPIO, I2C, UART, and other standard interfaces. Among all these requirements and based on the cost performance of different chips, Raspberry Pi 3b [9] becomes the first option.

As shown in Figure 11, the Raspberry Pi features a 1.4GHz 64-bit quad-core processor and 1GB of RAM. It is equipped with HDMI, a 2.4GHz Bluetooth module, and a 5G Wi-Fi module, which enables remote control capabilities. The board also includes a camera interface for connecting an external vision camera, which can serve as a source of environmental data for the robot. In addition, it offers a 40-pin GPIO header for interfacing with external circuits and outputting control signals.

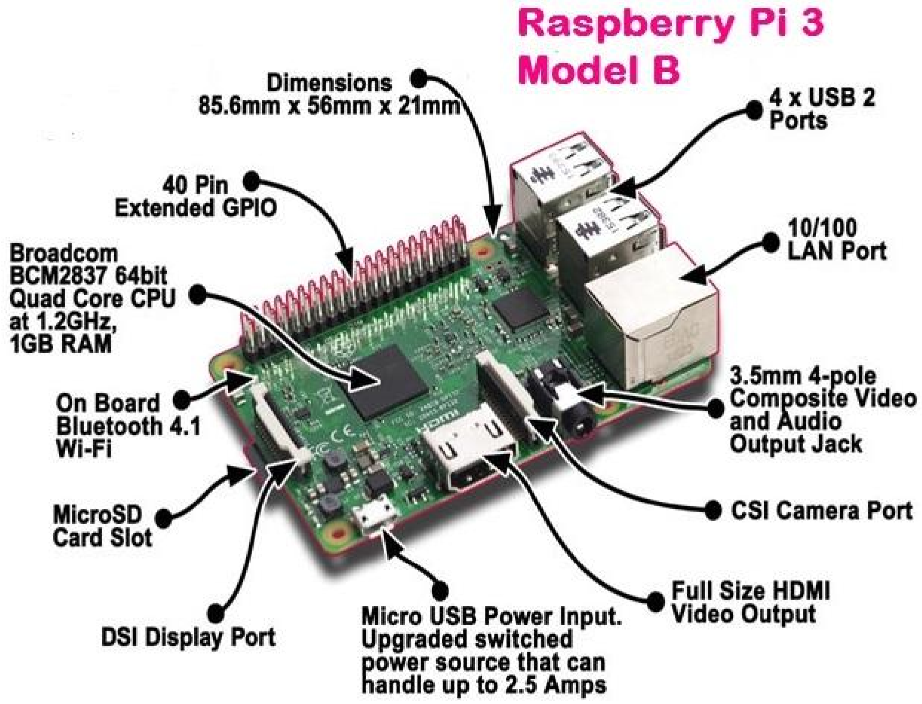

Figure 11. Rasperry Pi 3b

2.2.2. The Motor Drive Module

The motor drive module involves servo driver board, PCA9685 [10](Figure 12), and waterproof servo motor, Hitex HS-646WP(Figure 13). In which, the servo driver board communicates and receives messages from the main control chip through 12 C. It also outputs 16-channel PWM signals to control servo movements. Hitex HS-646WP waterproof servo has a maximum operating voltage of 7.4V. It offers a speed of 60°/0.17s and a maximum torque of 11.6 kg·cm, while weighing only 60g. With a compact size and an IP67 protection rating, it meets the experimental requirements of this study.

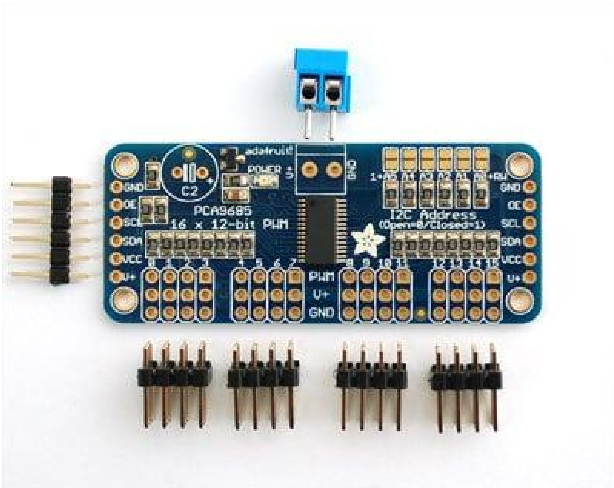

Figure 12. PCA9685 [10] servo driving board

Figure 13. Hitex HS-646WP waterproof servo motor

2.2.3. External Sensing Module

The external sensing module is displayed in Figure 14, which is applied on the bionic beaver [8] amphibious robot. The module uses high-precision waterproof ultrasonic distance measuring module and a high-precision attitude sensor from Beiwei [12] in order to obtain accurate orientation and motion data. In addition, the ultrasonic sensor applies enclosed modular waterproof design with an IP67 protection rating and exquisite structure, which involves a built-in measurement data stabilization algorithm, therefore having a relatively high output accuracy. In Figure 15, the 'Beiwei [12]' attitude sensor is a high-precision sensor with an angular resolution of 0.01°/s. It operates within a voltage range of 9–35V and features an IP67 protection rating, making it suitable for complex underwater operations. The sensor enables real-time measurement of the robot's orientation data.

Figure 14. Display of Ultrasonic Sensor from taobao.com

Figure 15. High Precision Altitude Sensor from Beiwei [12]

2.3. The Motions and Algorithms of The Bionic Beaver Amphibious Robot

2.3.1. Motion Analysis

2.3.1.1. The Swimming Motion Analysis of the Beaver

There are two conditions while the beaver [8] is swimming, which is the power stroke phase and the recovery phase. When the beaver [8] is in power stroke phase, all the joints on the hind limbs, which include hip, knee, and ankle, are all storing energy, thus the general posture is approaching the main body (Figure 16(a)). When the joints of the beaver [8] are pulling backward, the three joints of the hind limbs start to stretch out, providing a backward thrust. While stretching out, the water current will provide a counteracting force toward the flippers which makes the robot swim forward, the whole process is shown in Figure 16(b-d). When the beaver [8]'s hind limb is in motion shown in Figure 16(d). In the recovery phase, all of its three joints are all completely stretched out, which is shown in Figure 16(h). Therefore, when it is in the recovery phase, all of its three joints start to contract, and it helps to minimize the resistance of the water current, the whole alteration is presented in Figure 16(h-e). During swimming, a beaver [8] keeps its forepaws curled against its chest, maintaining a streamlined body posture to reduce water resistance and enhance swimming speed.

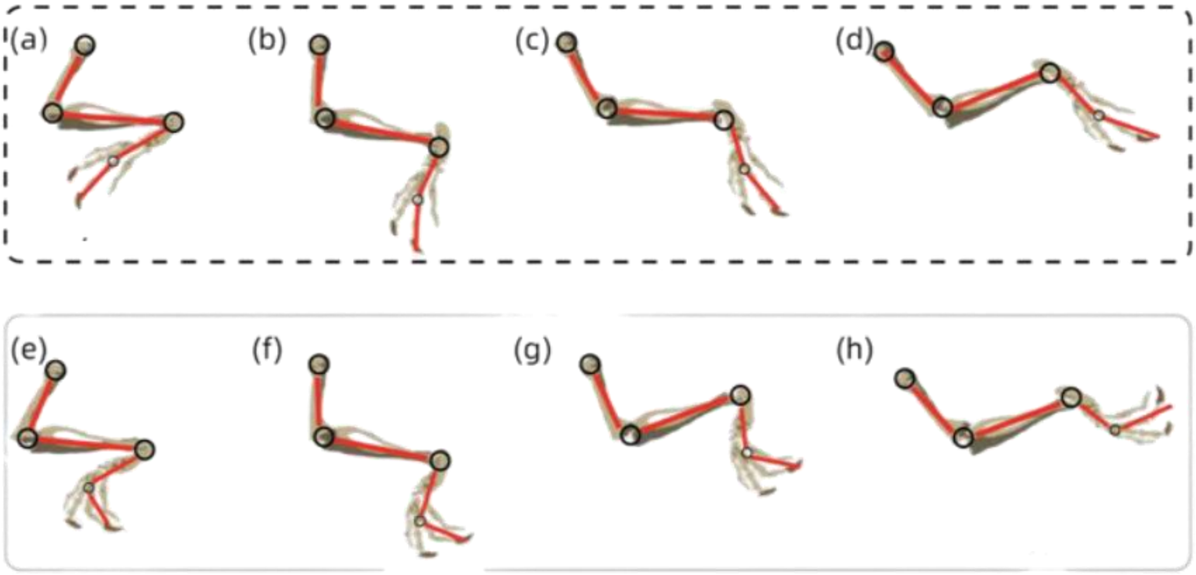

Figure 16. (a-d):When paddling backward, (e-h):When pulling the legs forward

During the beaver [8]'s swimming process, the backward paddling phase is defined as ψ, representing the beaver [8]'s paddling (propulsion) speed. The leg retraction phase is defined as φ, representing the speed at which the beaver [8] retracts its legs forward. The relationship between the two phases is shown in Equation (1).

$$\psi + \varphi = 1 \tag{1}$$

2.3.2. Inverse Kinematics of the Bionic Beaver Amphibious Robot

Based on the motion coordinates and analysis of the bionic beaver [8] amphibious robot, its inverse kinematics can be resolved by establishing sketches regarding the geometries of beaver [8]'s movements, for instance Fig 3.3. In Figure 3.5, the joint angles of the robot's hip, knee, and ankle are denoted as θhip, θknee, and θankle, respectively. The lengths of the robot's thigh, shank (lower leg), and flipper are denoted as l_1, l_2, and l_3, respectively. During the backward paddling phase, the flipper length l_3 is extended, while in the leg retraction phase, it is retracted and bent.

Based on the position {x, y} of the ankle joint end relative to the hip joint coordinate frame, the analytical inverse kinematics solution for the beaver [8]-inspired swimming robot is given by Equations (2) to (7).

$$L = \sqrt{|x|^2 + |y|^2} \tag{2}$$

$$\theta_{\text{knee}} = \arccos\left(\frac{l_1^2 + l_2^2 - L^2}{2 l_1 l_2}\right) \tag{3}$$

$$\alpha = \arccos\left(\frac{l_1^2 + L^2 - l_2^2}{2 l_1 L}\right) \tag{4}$$

$$\beta = \arctan\left(\frac{|y|}{|x|}\right) \tag{5}$$

$$\theta_{\text{hip}} = \pi - \alpha - \beta \tag{6}$$

A cubic polynomial fitting is used to model the proportional relationship between the ankle joint angle (θankle) and the knee joint angle (θknee). The ankle joint angle is then calculated using the fitted cubic polynomial function k, as shown in Equation (7).

$$\theta_{\text{ankle}} = k\theta_{\text{knee}} \tag{7}$$

2.3.3. PPO Algorithms

By applying the PPO [1][3][4] Algorithms, Proximal Policy Optimization, the bionic beaver [8] amphibious robot is capable of improving its locomotion constantly through the training process. The PPO [1][3][4] Algorithm is selected because of its balance between learning efficiency and training stability through multiple successive tasks. While using PPO [1][3][4], the neural network will receive comprehensive input vectors, which includes the robot joints' angles, body posture, and environmental feedback, such as water resistance. These inputs are all first processed through the multilayer perceptions to make the network capable of making decisions on the locomotive improvement of the robot.

The PPO [1][3][4] algorithm is operated with an actor critic structure, which has two branches, the actor branch and the critic branch. The former branch guarantees that the generated actions stay within physical and mechanical constraints, and the latter branch improves the robustness of training by providing accurate feedback to the policy updates. Thus, the truncated objective function of the actor branch is maximized and the mean square error of the critical branch is minimized to ensure the stability of training.

Combining the environment types indicators and the attention mechanisms, the amphibious robot is capable of switching to different motion modes depending on whether it's on land or in water. The resulting control allows the robot to perform coordinated and efficient paddling and pulling movements for both forward or backward propulsion. Its neural network diagram is shown in Figure 17.

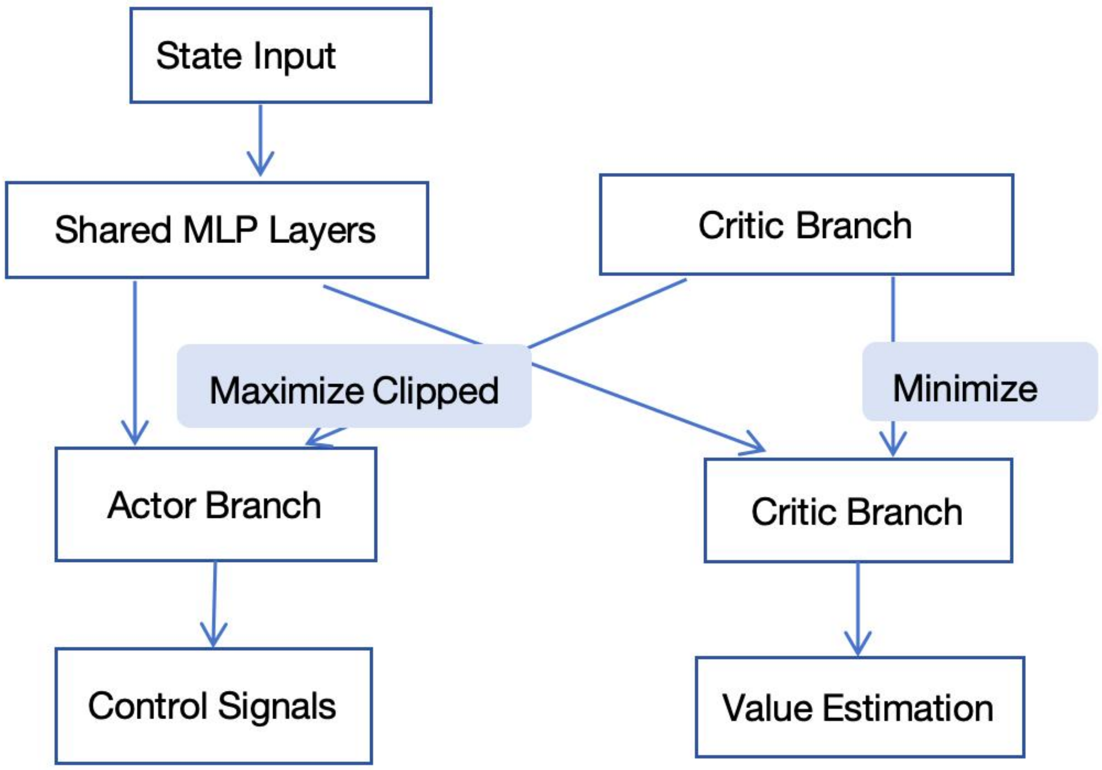

Figure 17. Neural Network Diagram

2.3.4. *The Prototype Testing*

Figure 18 shows the training process for a bionic amphibious beaver [8] robot based on the PPO [1][3][4] reinforcement learning [1][2][3][4][5] algorithm. On the Issac Gym simulation platform, the legged_gym framework was used to train the quadruped robot for both land and underwater locomotion.

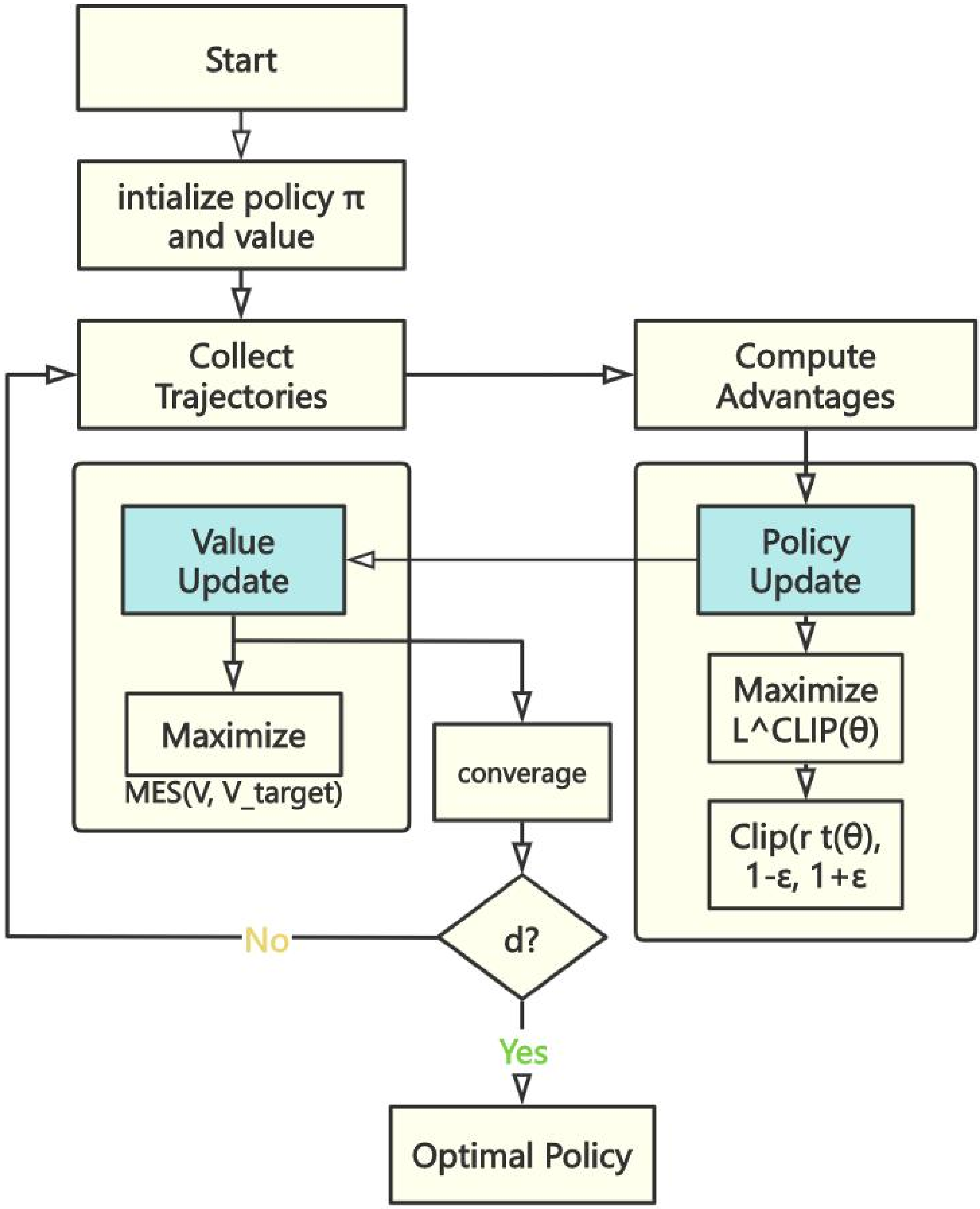

Figure 18. Flowchart based on PPO reinforcement learning

By adjusting the physical parameters shown in Tables 1 and 2, including gravitational acceleration, ground friction, and buoyancy coefficient, simulations of both land and underwater environments were achieved.

Table 1. Land simulation parameter settings （based on Isaac Gym)

Parameter	Value (Typical Value)	Description

Gravitational Acceleration	Vec3(0, 0, -9.8)	Z-axis upward
Ground Static Friction	0.8 - 1.2 (Concrete)	Ensure foot - end ground - gripping force and avoid slipping
Joint Damping	0.1 - 0.5 N·m·s/rad	Reduce energy loss and adapt to efficient terrestrial movement
Terrain Complexity	Plane / Slope (5° - 15°)	Use a flat surface in the initial training stage, and enhance robustness through domain randomization in the later stage.
Self - Collision Detection	self_collisions = 0	Enable self - collision (avoid leg penetration through the body)

Table 2. Water simulation parameter settings （based on Isaac Gym)

Parameter	Value (Typical Value)	Description
Water Density	1000 kg/m³	Standard water density
Buoyancy Coefficient	0.8 (Calculated Based on Volume)	F_buoyancy = ρ·g·V_submerged·0.8 (Simulate Partial Submersion)
Fluid Resistance	c_d = 1.2 (Trunk)	Drag formula: F_drag = 0.5ρC_dAv², C_d of legs = 0.6 (Reduce Water - paddling Resistance)
Joint Damping	2 - 5 N·m·s/rad	Increase damping to simulate water flow obstruction
Gravity Correction	Vec3(0, 0, -2.0)	Reduce effective gravity to simulate buoyancy balance in water

Through core reward design and some constraint behaviors, the motion models of land walking and swimming were simulated. Experimental tests were conducted on land and underwater, respectively, as shown in Figures 19-21 and Figure 22-24. To further test swimming capabilities, further outdoor testing was conducted, as shown in Figure 25. Figure 20 and Figure 23 shows the data regarding base velocity X, base velocity Y, base velocity yaw, the DOF position, the joint Velocity, Base velocity z, vertical contact forces, torque-velocity curves and joint torque through the experiment. In the graphs, the command vs. measured signals displays the learning process for the robot to follow target velocities. Morever, the fluctuations indicate the early training period whuch the agent is still exploring and exploiting the new skills giving by the command. The contact forces suggest the robot is beiging to establish a walking gait. Additionally, the yaw instability demonstrates the largest challenge the robot is facing while learning the mandatory skills it needs to learn.

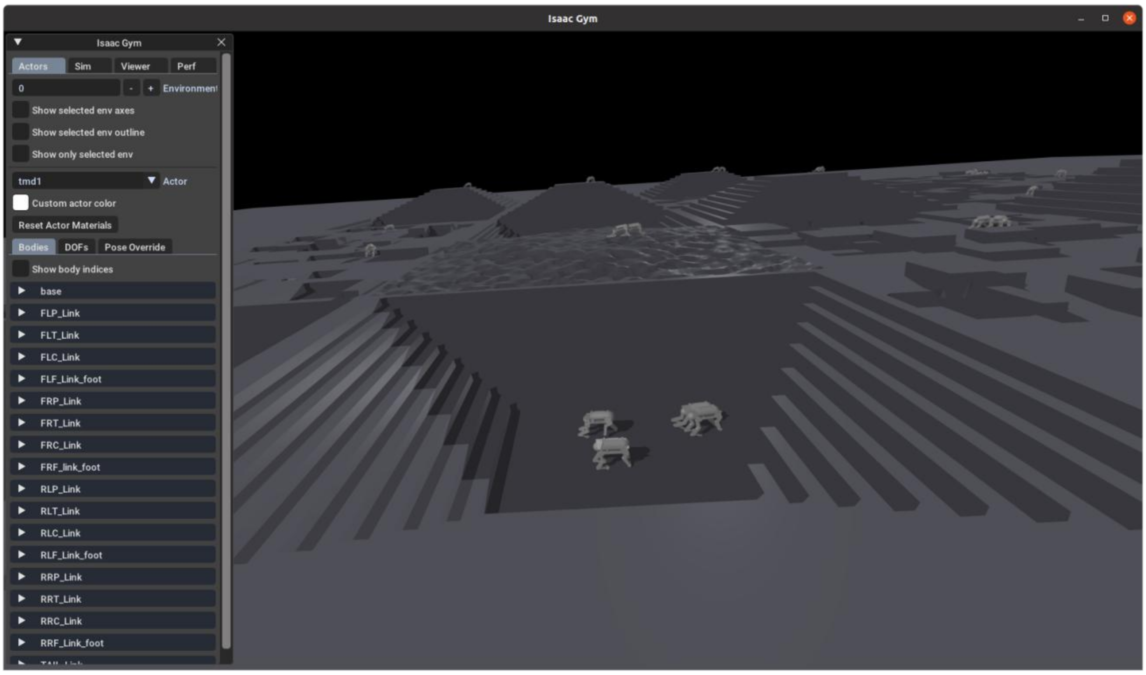

Figure 19. Issac Gym Terrestrial Locomotion Training

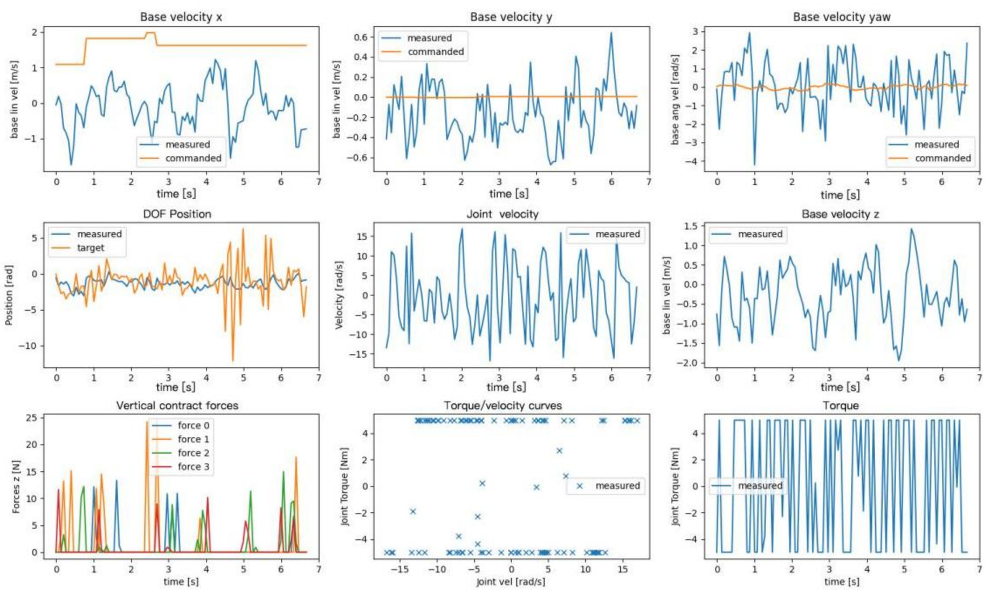

Figure 20. Issac Gym Training Result of Terrestrial Motion

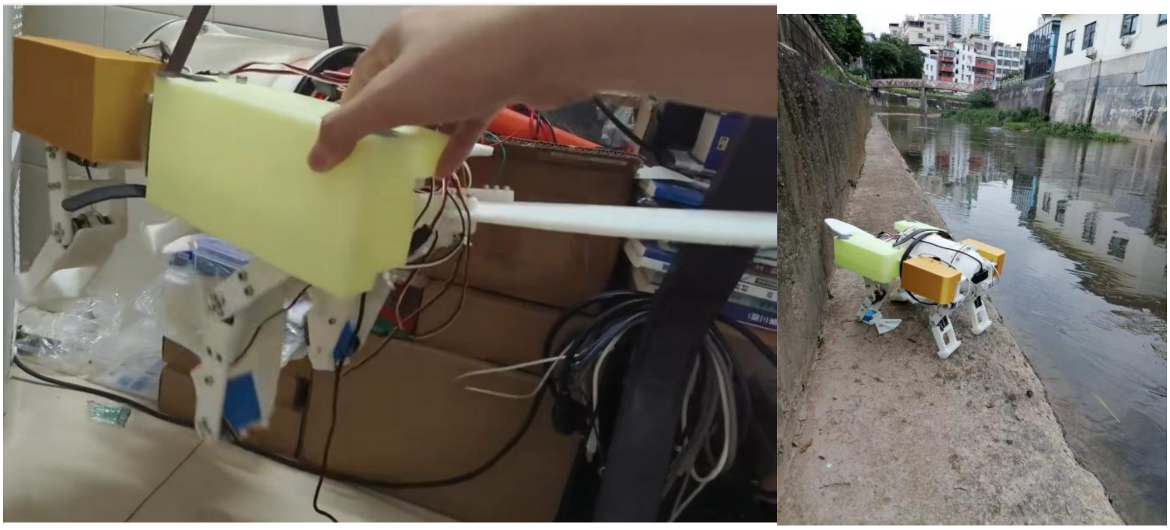

Figure 21. The Suspending Test and The Robot on Land

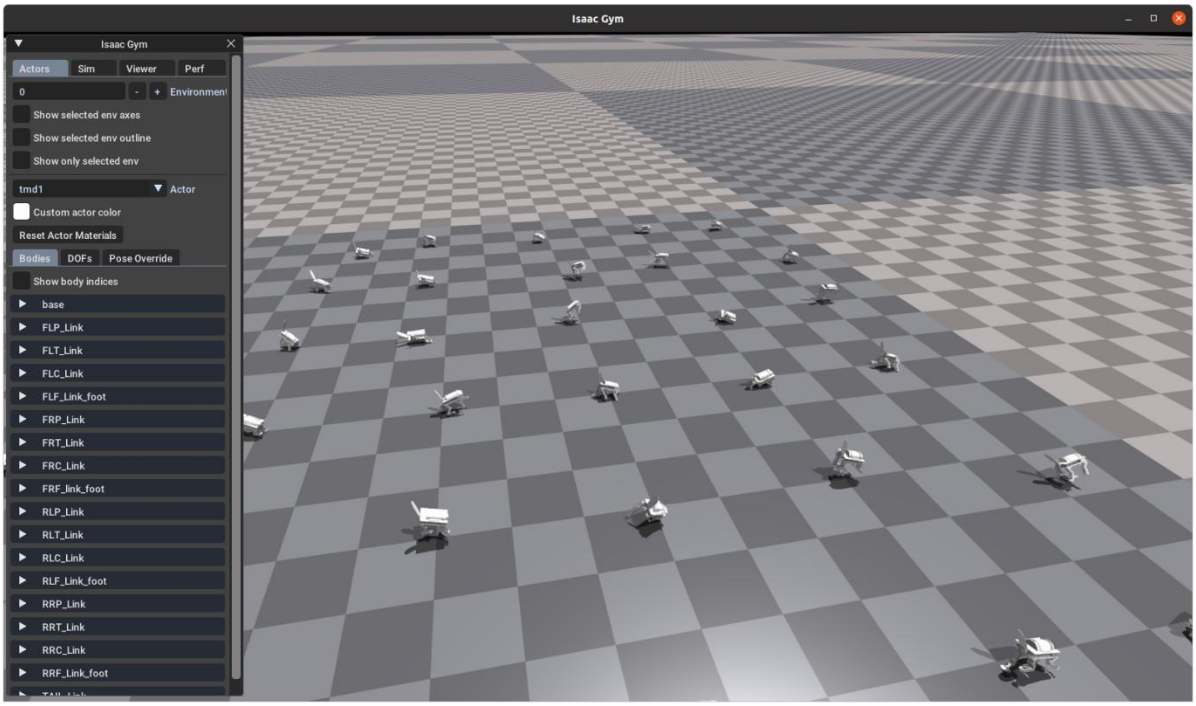

Figure 22.   The Aquatic Locomotion Training in Issac Gym

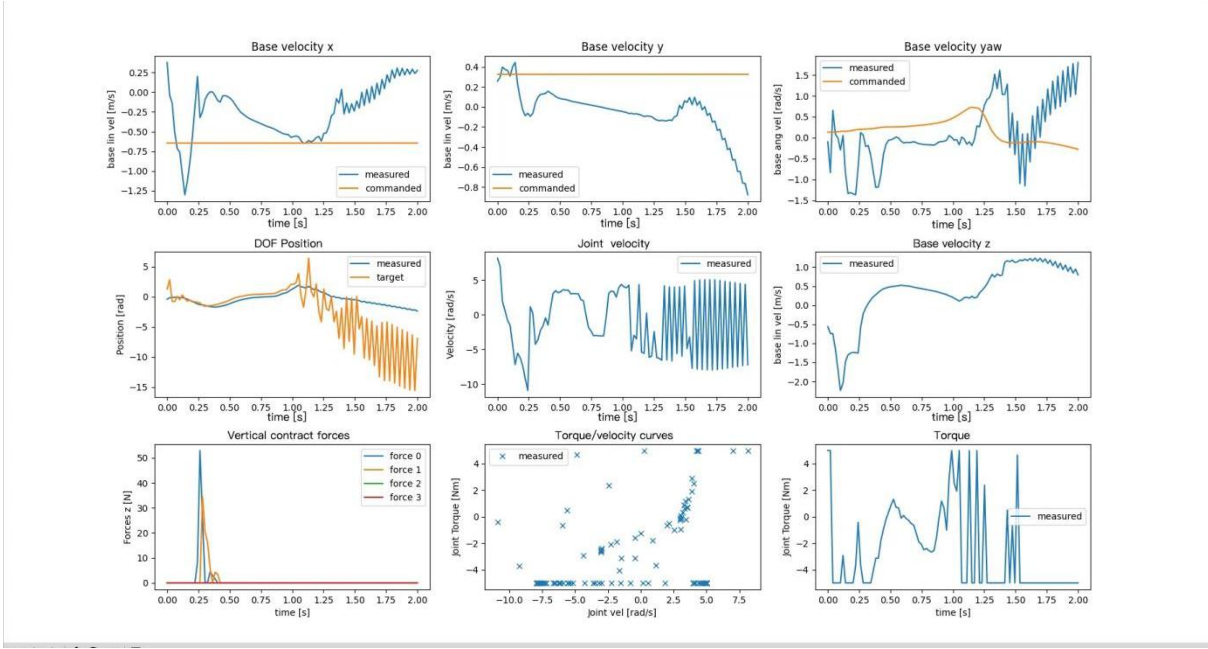

Figure 23. Aquatic Locomotion Testing Results

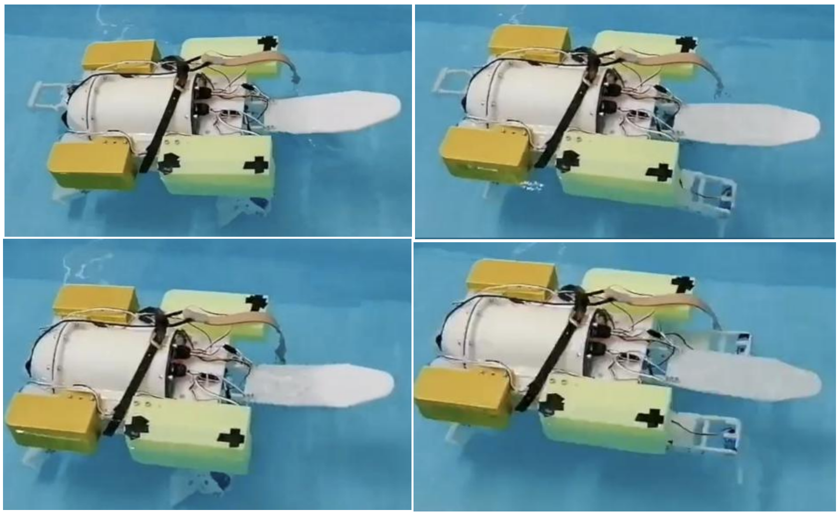

Figure 24. Real Time Robot Aquatic Locomotion Test

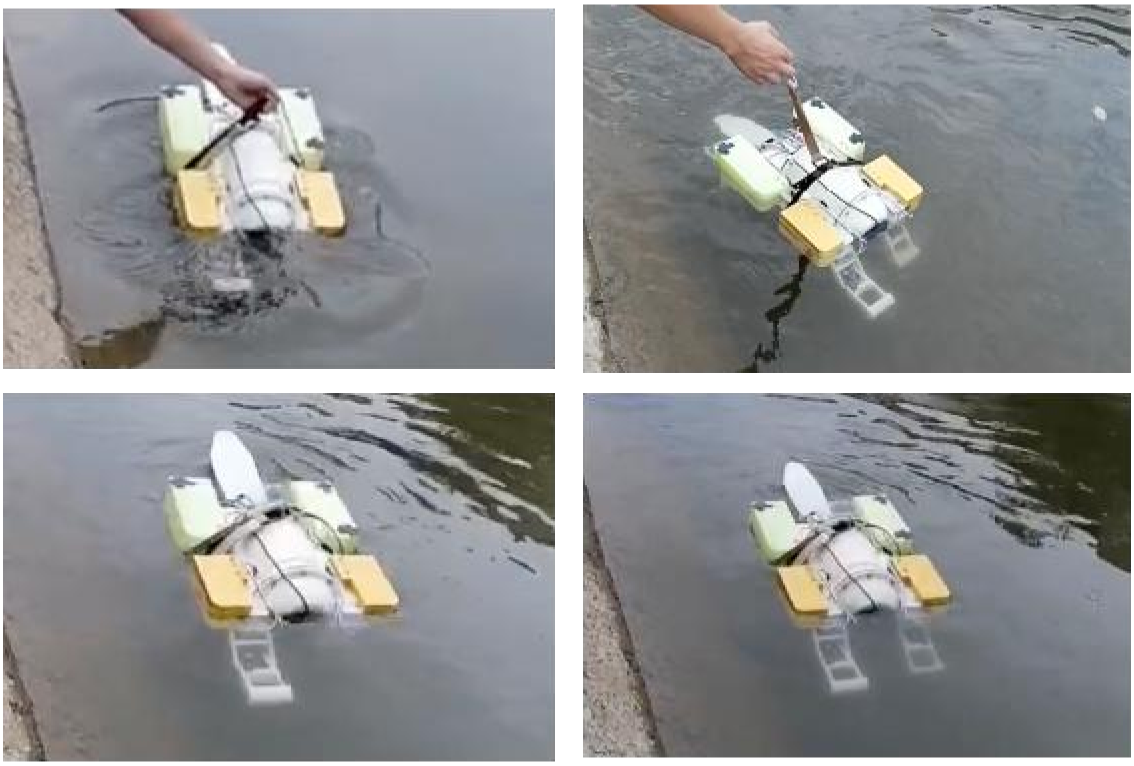

Figure 25. Outdoor swimming test

2.4. Functionality Comparisons

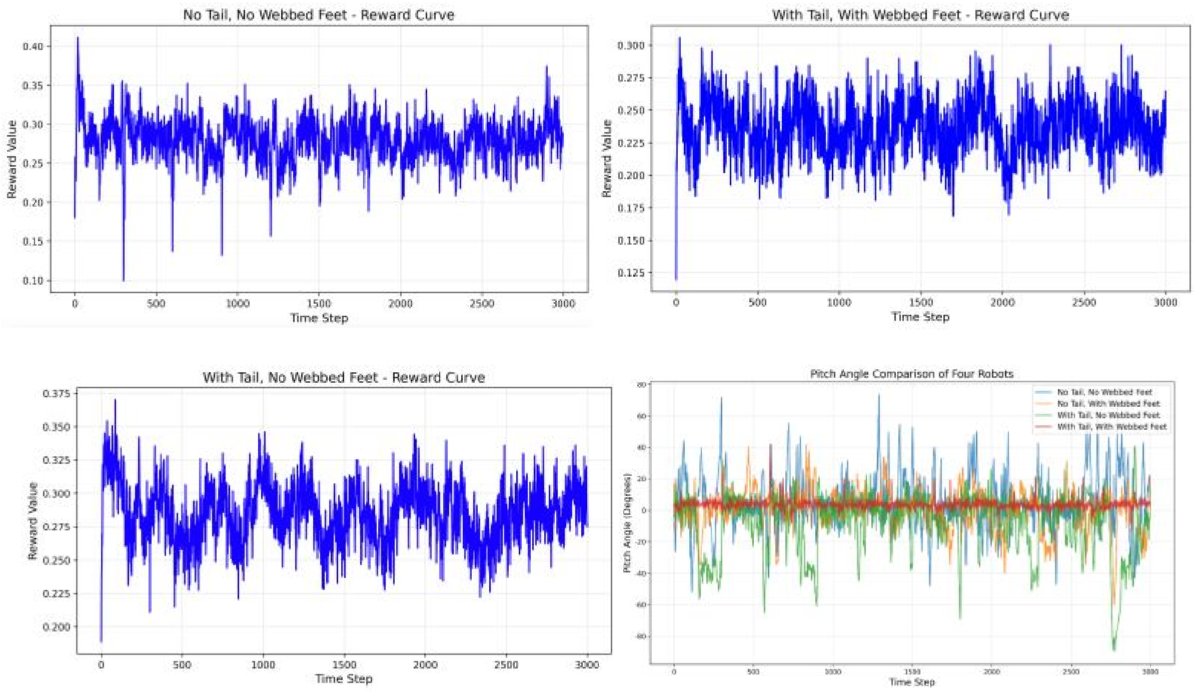

Figure 26. Reward Value Curves under Different Structures and the Robot's Pitch Angle

In a series of synchronized swimming experiments conducted by using the PPO [1][3][4] algorithms, the performance of the bionic beaver [8] amphibious robot was evaluated under 4 distinct morphological configurations. No tail and no webbed feet, with tail and webbed feet, with tail and no webbed feet, and no tail but with webbed feet. In Figure 26, it shows the changing trends of the reward curves, which serve as an indicator of the robots' learning efficiency and the locomotion effectiveness in water. The pitch angle variation during training is also shown in the figure. Among all the configurations, the setup that involves both webbed feet and tails demonstrate the most stable and consistent high reward values, which means it provided the most advantageous morphology for underwater propulsion and control.

Additionally, the bottom-right plot of Figure 26 compares the pitch angle behavior across the four configurations through the training. The results reveal that the robot equipped with both a tail and webbed feet maintained a narrow pitch angle range throughout training, exhibiting strong and consistent stability. On the other hand, the configurations lacking a tail or webbed feet showed greater pitch variation and erratic angular responses, likely due to insufficient balance and hydrodynamic resistance. These results all show that having a tail and webbed feet not only enhance the robotic propulsion but also contributes to a more stable and efficient body posture regulation. Thus, the robot is able to learn and maintain a well-setup swimming locomotion, adapting well to the aquatic environment, and showcasing the importance of bioinspired structural features in amphibious robot design.

3. Discussions

By integrating the bionic mechanical design with the PPO [1][3][4] algorithm, the bionic beaver [8] amphibious robot is able to switch between different locomotions depending on diverse environments. Generally, the design of the tail and the webbed feet modelled the morphology of the beaver [8]. The tail is capable of passive bending, which advances the stabilization of the robot while swimming. In addition, by coupling with the bipedal limbs and the webbed foot design, the robot achieves enhanced propulsion efficiency, improved maneuverability, and increased adaptability to turbulent and unstructured aquatic environments.

Additionally, by comparing the functionality of the bionic robot with or without either the flippers or the tail, the results of the comparison reveal that the pitch variation and erratic angular responses are minimized by having a tail and flippers. Therefore, tails and flippers will provide strong and consistent stability while the robot is swimming in water.

A key innovation in this robot is the application of PPO [1][3][4] reinforcement learning [1][2][3][4][5]. Unlike other traditional controllers, such as manually-tuned controllers or high-fidelity computational fluid dynamics (CFD) models, the PPO [1][3][4] algorithm enables the robot to autonomously learn effective motion strategies through direct interaction with a simulated environment. This data driven approach greatly reduces the modeling complexity and maintains robust performance.

This hybrid innovation of combining the bionic mechanical structure with PPO [1][3][4] allows the robot to maintain stable posture and constant propulsion under dynamically changing flow conditions, which ensures its functionality in real world conditions. Moreover, the learning policy illustrates the generalization across different configurations, therefore the control framework can accommodate any changes of structural variations, such as fin shape, limb length, or joint flexibility, etc.

Overall, this combined design that involves bionic mechanical design and PPO [1][3][4] algorithms has effectively advanced the locomotive stability and efficiency of bionic amphibious robots, which indicates the future potential and further improvement needed in the field of underwater robots. This research also validates the effectiveness of bionic robotics and demonstrates the potential of reinforcement learning [1][2][3][4][5] as an effective and economical tool for locomotion.

4. Conclusion

Generally, this research presents the mechanical and control design of the bionic beaver [8] amphibious robot, which has fin and bipedal limbs equipped with multi degree of free joints. This design enables the robot to have an agile and stable locomotion in water. By applying the PPO [1][3][4] reinforcement learning [1][2][3][4][5], the robot achieves real time motion control without using any other computationally expensive fluid dynamic modeling, which ensures the robot can perform its optimal functionality in complex and variable flow conditions.

The experimental results show the effectiveness of this integrated design learning approach, which illustrates the robot's capability of adapting to different morphological configurations and maintaining its mechanical performance in various aquatic conditions. However, to further improve its performance and its application scope, several enhancements can be made. Structurally, increasing the external swing hip joint, increasing the snorkeling device, increasing the attitude control structure and other parts can make the robot achieve a more delicate bionic structure. On the control side, improvements in motion planning and precision, such as pitch angle regulation, which is essential for navigating more challenging underwater terrains.

Sensors can also be optimized. By upgrading and tuning the robot's sensory system, including inertia, depth, and flow sensors, it can greatly improve the performance and the adaptability of the robot in diverse aquatic environments. Additionally, applying Sim-to-real transfer techniques will be crucial to bridge the gap between simulation and real time application. These methods enhance the robustness of the trained policies and elevate the adaptations when transferring the robot from simulated environment to real aquatic environment, such as pools, rivers, or lakes, etc.

Finally, this research highlights the feasibility and the advantages of integrating bionic mechanical design with reinforcement learning [1][2][3][4][5] for amphibious robots, while also highlighting a promising direction for underwater robots. The proposed enhancements in morphology, control accuracy, sensing, and Sim2Real transfer will collectively enable the development of more robust, adaptive, and capable amphibious robots for real-world applications, such as environmental monitoring, exploration, and rescuing operations in aquatic environments.

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